Final Exam Topics (Front)

Anything I cover in class is fair game for the final. To help you here is a partial list of some of the items that we covered in class.

C++ Basics

- Machine Language, Assembly Language, High-Level Language
- Namespace and scope resolution operator
- `using namespace std`
- Fundamental data types: bool, char, int, float, double…
- Declaring and initializing variables
- C control structures: if, else, while, do, for, switch, break
- C operators: +, -, *, /, --, ++, >>, <<,&&, ||, ^, %...
- Type conversions and typecasting
- Preprocessor directives: #include, #define, and inclusion guards
- I/O using cout, cin
- I/O stream manipulators: setw, setfill, hex, oct, dec, fixed, scientific, setprecision
- File I/O: sequential text file manipulation
- Post- and pre-increment and decrement
- Header files vs source files
- Global and local variables, variable scope rules
- Storage class – automatic vs. static
- Standard c libraries (e.g. #include<cmath>)
- Standard math library functions: ceil, floor, fabs, exp, log, cos,sin, sqrt, tan, rand…
- Functions, function prototypes, argument coercion, overloading
- Default arguments in functions
- Pass-by-value and pass-by-reference
- Recursion (Fibonacci vs. Factorial)
- Order of precedence
- Mixed expressions, integer division
- Arrays: size, initialization, passing to function
- Using function templates
- Pointers: declarations, initialization, dereferencing, arithmetic, arrays, strings, functions,
- Address operator &, dereferencing operator *, member access operator ->
- References vs. pointers
- Use of `const` keyword
- Know `fstream` objects (ifstream and ofstream) – sequential text file manipulation
- C++ string objects vs. pointer base strings

Classes

- Constructors and Destructors – when are they used and how they are written
- Overloading constructors, initializer lists. etc…
- Class composition
- Member functions: public vs. private
- Set and get member functions
- Data members: Public and Private
- Friend functions and classes, const and static class data members and functions
- Use of the "this" pointer in classes
- Operator overloading – member function vs. global operator function
- Function templates and class templates
- Using a `vector and list`
- STL – containers, iterators, and algorithms

Final Exam Topics (Back)

Dynamic Memory Management

C++ memory management features – new and delete operators
Using dynamic memory management in class objects
The rule of three
Linked List and Dynamic Arrays

Inheritance

Using a base class and a derived class
Constructors, destructors, and member functions
Inheritance constructor calls in initializer list
Public inheritance
Protected data members

Polymorphism

Pointing base pointers (or references) at derived classes
Using derived classes through a base reference
Virtual and pure virtual functions
Polymorphic processing
Polymorphic data structures (i.e. dynamic array of base pointers or reference)
Polymorphic functions (i.e. passing base pointer or base reference)

Using C++ and C strings and string functions

Ability to use the member functions and operators
You will have to write some code to manipulate strings:
`size()`; `find ()`; `rfind ()`; `find_first_of ()`; `find_last_of ()`; `find_first_not_of ()`;
`c_str()`; `substr()`; `erase()`; `insert()`;

Data Structures Manipulation

Linked List Manipulation at the Pointer Level
Dynamically Allocated Arrays
Stacks
`vector` in STL, `list` in STL

Standard Template Library

range-based for loops
map
vector
list
sort algorithm

Multithreaded Computing

launching threads
join()
detach()
mutex objects (lock() and unlocked())
resource contention
race conditions